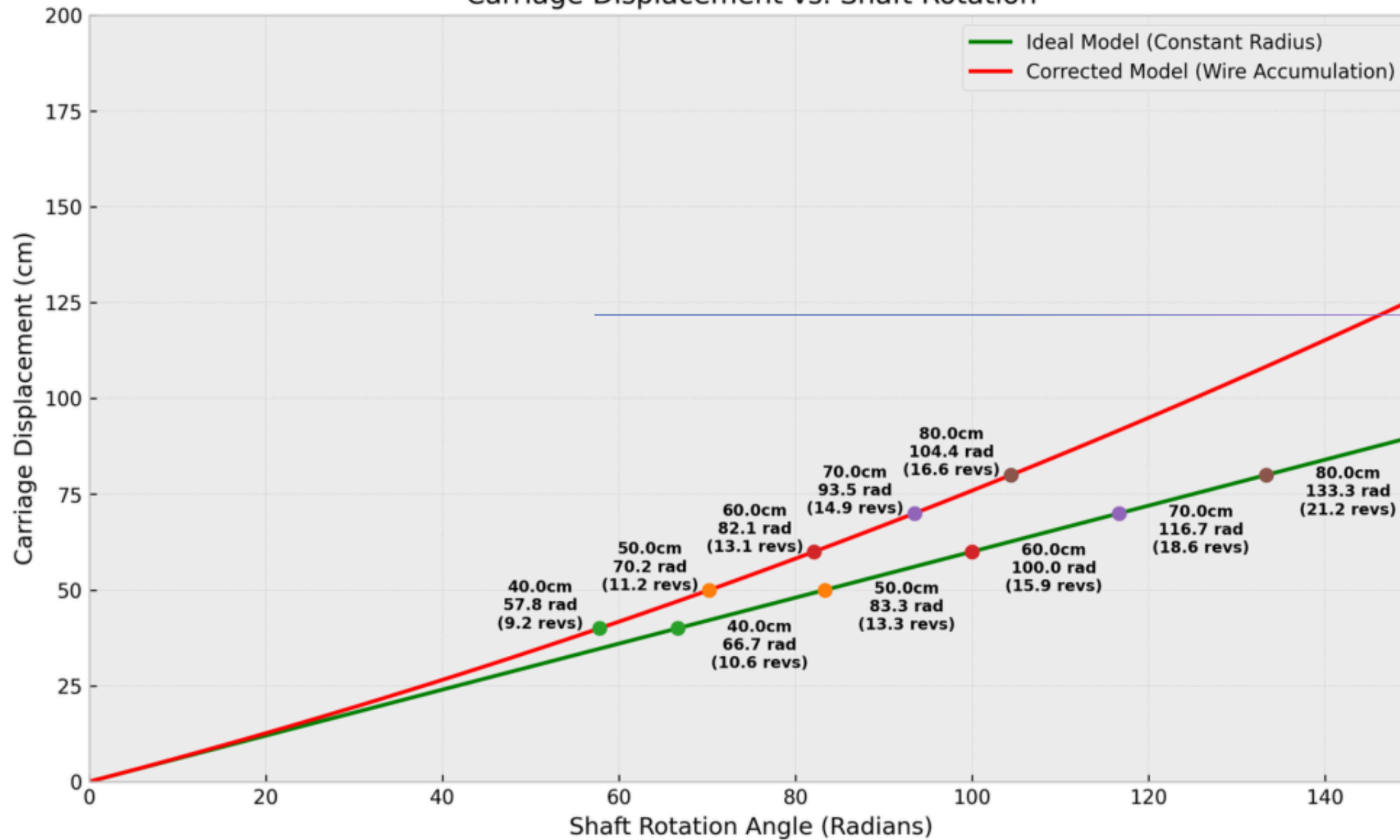


CaDDE

Carriage Displacement vs. Shaft Rotation



Wire radius (**0.1mm**)

Idea Model:

Constant radius

Wire accumulation model:

Effective radius grows

$$r_{eff} = r_{shaft} + \left(r_{initial} \times \left(\frac{\theta}{\pi} \right) \right)$$

DC Motor chosen: 12V 18RPM w/ Encoder

No Load RPM: **18** ±10%

Rated Load Operating Current: 0.2A

Rated Load RPM: **13** ±10%

Rated Load Torque: 12kg (**117.7N**)

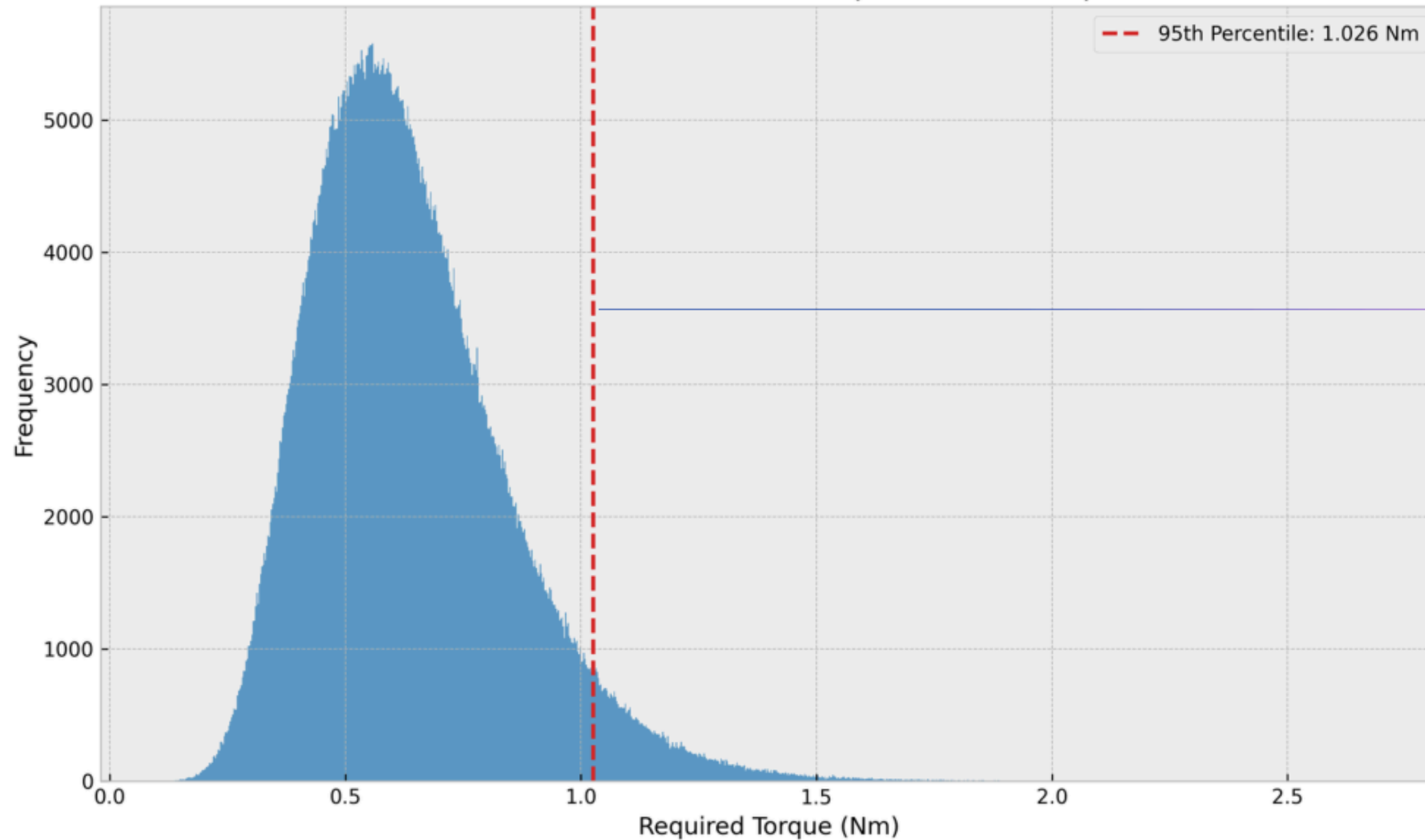
Rated Load Power: 2.4W

Stall Torque: 34kg (**333.5N**)

Stall Current: 1.3A

CaDDE

Monte Carlo: Distribution of Required Motor Torque



User mass: Normal distribution

(Mean - 60kg, Standard Deviation - 10kg)

Friction Coefficient: log normal distribution

(Mean - 0.18, Standard Deviation - 0.05)

Shaft Radius: Normal distribution

(Mean - 0.006m, Standard Deviation - 0.0003m)

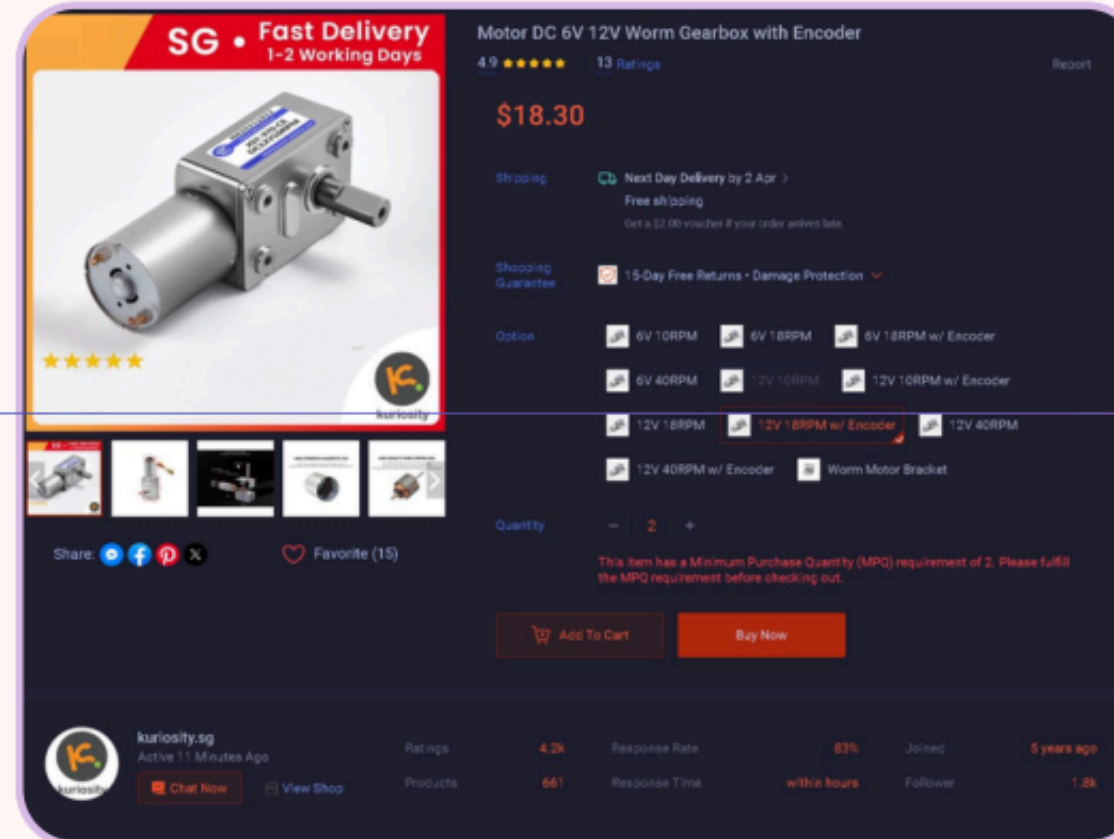
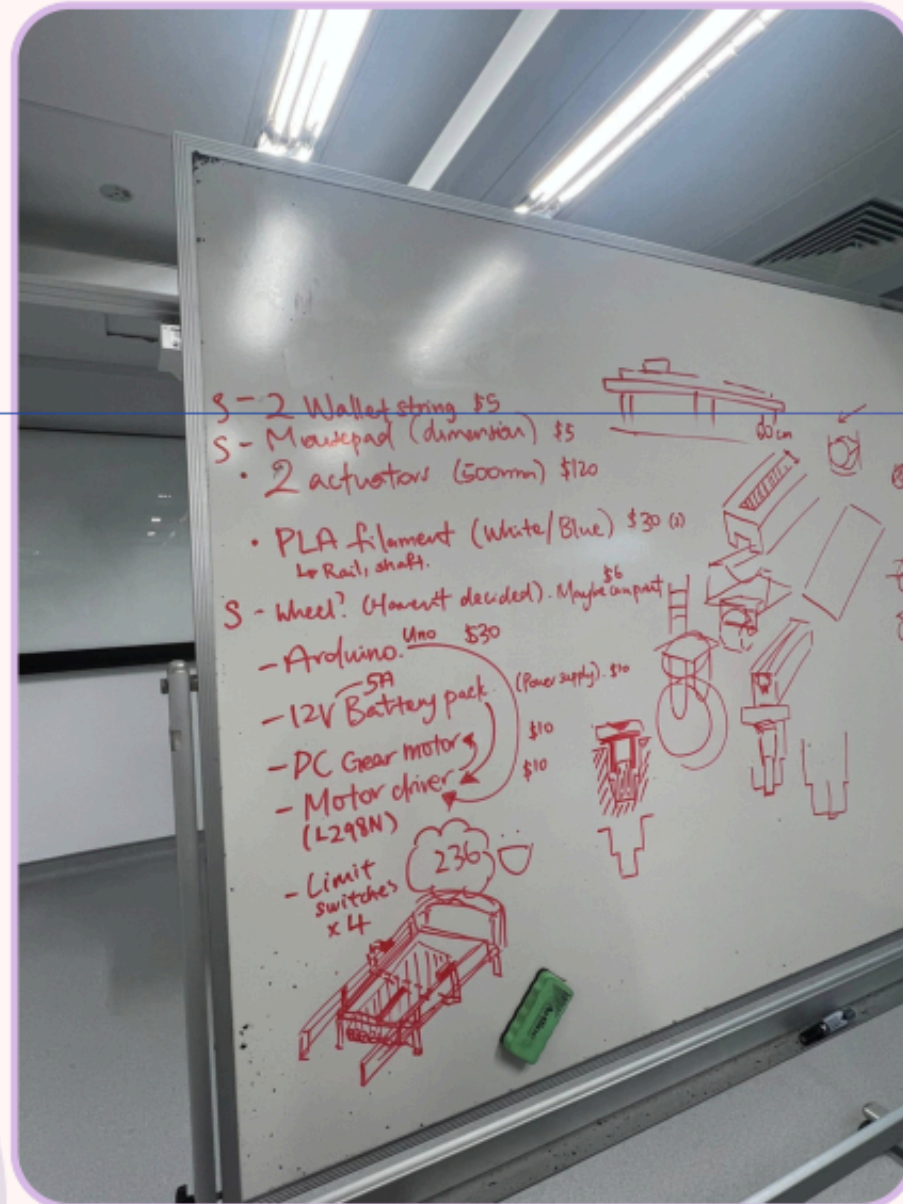
$$T = \mu \times m \times g \times r$$

Based on **1,000,000** simulations, we have determined that our motor must provide at least **1.026 Nm** of torque to reliably assist 95% of our target demographic.

With a safety factor of 2, we require **2.052Nm total** motor torque.

Using a dual motor setup, each motor needs **1.026Nm** normal operating torque

Purchasing Logistics



Confirming using online stores on what models are available

Items purchased from Sim Lim and online!

Confirming prototype and brainstorming corresponding logistics

Pending the arrival of the shaft